

## ⚡ PowerPulse – Household Energy Usage Forecast & Analysis

### ✦ Project Overview

PowerPulse is a comprehensive data science and machine learning project focused on household electricity consumption analysis, forecasting, peak usage detection, and anomaly identification using real-world smart meter data.

The project leverages historical power consumption data to:

Predict household energy usage

Identify peak demand periods

Detect anomalous consumption behavior

Provide actionable insights for cost optimization and energy efficiency

This repository demonstrates end-to-end data analytics, including EDA, feature engineering, regression modeling, time-based analysis, and anomaly detection, making it suitable for academic projects, interviews, and real-world energy analytics use cases.

### 📁 Repository Structure

PowerPulse-Household-Energy-Usage-Forecast/

```
|
|
|— data/
|   └─ household_power_consumption.txt
|
|
|— notebooks/
|   └─ Energy_Consumption_Forecasting.ipynb
|   └─ Peak_Usage_Analysis.ipynb
|   └─ Time_Based_Energy_Analysis.ipynb
```

```
|
|— scripts/
|   |— forecasting_models.py
|   |— peak_detection.py
|   └— anomaly_detection.py
|
|— results/
|   |— model_performance.png
|   |— feature_importance.png
|   └— consumption_patterns.png
|
|— README.md
└— requirements.txt
```

## Dataset Information

Source: UCI Machine Learning Repository

Dataset: Individual Household Electric Power Consumption

Time Period: 2006 – 2010

Frequency: Minute-level readings

### Key Features

Global\_active\_power

Global\_reactive\_power

Voltage

Global\_intensity

Sub\_metering\_1 (Kitchen)

Sub\_metering\_2 (Laundry)

Sub\_metering\_3 (Climate systems)

Date, Time

## ☐ **Project Components**

### 1 ☐ **Energy Consumption Forecasting (Machine Learning)**

#### **Objective**

Predict Global Active Power consumption using historical usage patterns and engineered time-based features.

#### **Key Steps**

Missing value handling using forward & backward filling

Skewness correction using logarithmic transformations

#### **Feature engineering:**

Hour, day, month, weekend flag

Daily mean, max, and min consumption

## Model training & evaluation

### Models Implemented

Linear Regression

Random Forest Regressor

Gradient Boosting Regressor

Neural Network (MLP)

### Evaluation Metrics

RMSE

MAE

R<sup>2</sup> Score

### Results

| Model               | RMSE  | MAE   | R <sup>2</sup> |
|---------------------|-------|-------|----------------|
| Linear Regression   | 0.548 | 0.378 | 0.728          |
| Random Forest       | 0.402 | 0.249 | 0.854          |
| Gradient Boosting ★ | 0.392 | 0.240 | 0.861          |
| Neural Network      | 0.393 | 0.245 | 0.861          |

✓ Best Model: Gradient Boosting Regressor

Feature Importance Insights

Sub\_metering\_3 (Climate systems)

Sub\_metering\_1 (Kitchen)

Hour of the day

Daily mean consumption

## 2 Peak Power Usage Analysis

### Objective

Identify daily peak energy consumption periods to support load balancing and cost optimization.

### Methodology

Hourly aggregation of Global Active Power

### Dynamic peak threshold:

Peak Threshold = Mean + Standard Deviation

### Visualization of peak windows

🔥 Identified Peak Hours

19:00 – 22:00

## Key Findings

Metric Value

Average Consumption 1.09 kW

Peak Consumption 1.84 kW

Peak Hours per Day 3

Avg Peak Energy 5.51 kWh

These hours align with evening household activity, validating real-world behavior.

## 3 Time-Based Energy & Anomaly Analysis

### Objective

Analyze time-dependent energy usage, estimate cost savings, and detect abnormal consumption patterns.

### Features

Peak vs Off-Peak comparison

Weekend vs Weekday usage

Hourly consumption profiling

Cost simulation using time-based tariffs

Anomaly detection using Isolation Forest

### Cost Analysis (Sample Output)

Metric Value

Current Monthly Cost USD 421,359

Potential Savings USD 30,583

## **Anomaly Detection**

Model: Isolation Forest

Detected ~206,000 anomalous readings

Helps identify:

Faulty appliances

Power leakage

Unauthorized usage

## **Business Use Cases**

Smart home energy optimization

Utility demand forecasting

Dynamic pricing strategy support

Smart grid analytics

Sustainability and carbon footprint reduction

## **Technologies Used**

Language: Python

## **Libraries:**

Pandas, NumPy

Matplotlib, Seaborn

Scikit-learn

Models: Regression, Neural Networks, Isolation Forest

## **🔮 Future Enhancements**

LSTM-based time-series forecasting

Weather data integration

Real-time IoT streaming support

Interactive dashboard (Streamlit / Power BI)

Carbon emission estimation